Deliverable 3: Interim Status Report

Cybersecurity Analytics Class Project

Deliverable	Deliverable 3: Interim Progress Report 2	**Selected Track**	Fraud
Project Title	Deepfake Recognition	**Team Name**	Team 6
Team Members	Flemming Grabowski, Payel Sarkar	**Submission Date**	November 8, 2026
Repository	https://github.com/payelsarkarr/is665-team6-deepfake-recognition	**Dataset Link**	FaceForensics++ (official release, c23, access granted): https://github.com/ondyari/FaceForensics

Executive Status Snapshot

Item	Current status
Overall status	On track. Since Interim Report 1 we downloaded the full official dataset, rebuilt the pipeline in PyTorch, trained and tested the baseline and both planned models on all 5,000 videos, and tested generalization to unseen manipulation methods (Section 5).
Best-performing approach so far	MobileNetV2 (frame model): test ROC-AUC 0.967, recall 0.965 at a validation-chosen threshold. The GRU clip model is not significantly better (ROC-AUC 0.976, overlapping 95% CIs), does not generalize better to unseen methods, and costs about 6× the training time.
Most important error or risk	Unseen manipulation methods. With one method left out of training, recall on that method falls to 13–90% (FaceSwap: ROC-AUC 0.41, below chance). A fraudster using a new face-swap tool would likely pass. Second: about one in five real videos is flagged (test FPR 0.217).
Work remaining before final report	Robustness/adversarial experiment with a defense, slice analysis of false alarms, Grad-CAM, inference latency, demo, and repository hardening (Section 9).

1. Updated Project Status

Completed since Interim Report 1. The official FF++ servers came back on Oct 16. We downloaded all 5,000 c23 videos and extracted face clips for 4,999 (one has no detectable face), meeting the Oct 18 milestone on Oct 17. We then trained the baseline and two candidate models, tested them on the official test split, ran a leave-one-method-out (LOMO) experiment and inspected the worst errors.

Changes. (1) Official data only: the Kaggle copy and the 1,250-video subset are no longer used. (2) PyTorch instead of Keras, in a modular pipeline (src/ffpp; notebooks 01 prepare, 02 train, 03 evaluate) that saves each run's configuration and a checkpoint per epoch. (3) Clips resampled to about 25 fps, 4 clips × 16 frames per video, one fixed face box per clip. (4) LOMO added as the generalization test inside FF++.

Feedback on Interim Report 1 and how we addressed it.

- **Behind schedule / at risk:** the Oct 18 milestone was met, and the baseline and both planned models are trained and evaluated.
- **Unofficial copy and small subset:** all results now use the official release with all 5,000 videos; the Kaggle copy is no longer used.
- **Planned model not started, small validation set:** the CNN baseline, MobileNetV2 and MobileNetV2 + GRU are trained; validation grew from 101 videos (21 real) to 655 (135 real).
- **Uncertainty on a source/method-disjoint test:** the test split is source-disjoint (official splits plus removal of YouTube sources shared across splits), LOMO gives method-disjoint tests (Section 5), and all main results carry 95% bootstrap intervals.

Readiness: data, code and the evaluation pipeline are complete and in use.

2. Model Development and Comparison

A video's score is the mean P(fake) over 8 frames (or its clips). **Validation:** official FF++ splits (720/140/140 source videos; fakes follow their target video). Seven YouTube sources shared across splits are removed from validation and test, leaving 655 validation (135 real) and 682 test videos (138 real). The best epoch is chosen by validation ROC-AUC, the threshold on validation (highest recall with FPR ≤ 0.20); test was used once.

Tuning. No grid search (one Colab T4). Standard transfer-learning settings: AdamW, warm-up then cosine decay, frozen backbone in epoch 1, early stopping (patience 3); augmentation (flip, brightness, JPEG quality 40–90) identical for all frames of a sample. **Imbalance (1:4):** class-weighted loss and an FPR-based threshold instead of 0.5. **Reproducibility:** seed 665, config.json per run, cached extraction settings, code in the repository; one run per configuration, so uncertainty comes from bootstrap intervals.

Table 1. Approaches compared (official splits; T4 GPU).

Approach	Key settings	Validation method	Rationale / result
Baseline B2: small CNN from scratch (1.17 M parameters)	128 px frames, 4 conv blocks, lr 1e-3, 10 epochs, 0.8 min/epoch	Official train/val; best epoch by val ROC-AUC	Shows what a CNN learns without pretraining. Val ROC-AUC 0.70, still rising (underfit).
Model 1: MobileNetV2, ImageNet weights, per frame (2.23 M)	224 px, lr 3e-4, 1 frozen epoch, 8 epochs, 1.1 min/epoch	Same	Transfer learning, light enough for a demo. Val ROC-AUC 0.975.
Model 2: MobileNetV2 + bidirectional GRU over 16 frames (4.59 M)	224 px clips, lr 1e-4, 1 frozen epoch, 6 epochs, 7.0 min/epoch	Same	Tests whether temporal inconsistency adds evidence. Val ROC-AUC 0.977.

3. Preliminary Evaluation

Primary metric: recall at a false-positive rate of at most 0.20, with the threshold fixed on validation. It matches how the detector would be used: analysts can review a limited number of false alarms, and within that budget it should catch as many fakes as possible. ROC-AUC compares models independent of the threshold. Accuracy is not used: labelling every video fake already gives 80%. Precision and PR-AUC look high only because 80% of the test videos are fake (see the operational note).

Table 2. Test results at the validation threshold (n = 682 videos, 138 real). 95% bootstrap intervals (1,000 resamples) in brackets.

Model	Precision	Recall	F1	PR-AUC / ROC-AUC	Operational note
Baseline B2	0.867	0.467	0.607	0.884 / 0.684 [0.630, 0.733]	FPR 0.283. Misses half of the fakes.
Model 1: MobileNetV2	0.946	0.965	0.955	0.992 / 0.967 [0.952, 0.980]	FPR 0.217 [0.155, 0.294]; 30 false alarms, 19 misses. 2.2 M parameters.
Model 2: MobileNetV2 + GRU	0.940	0.976	0.958	0.994 / 0.976 [0.965, 0.986]	FPR 0.246 [0.177, 0.315]; 6× training time per epoch.

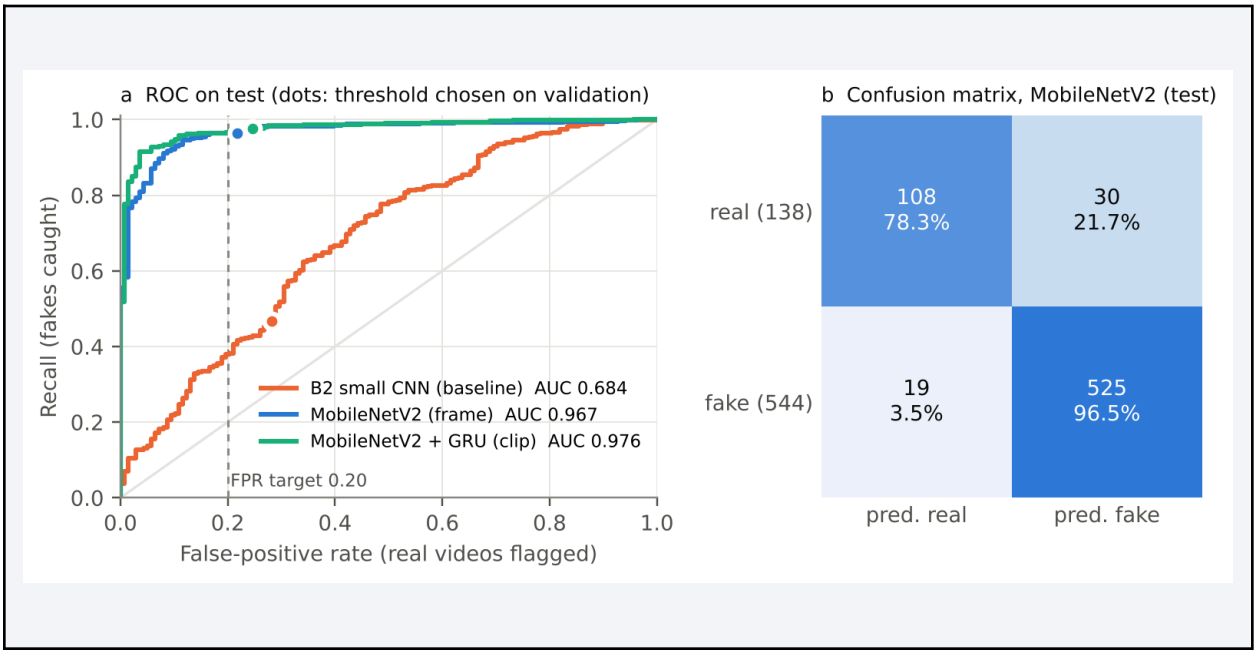


Figure 1. (a) Test ROC curves; dots mark the threshold chosen on validation. (b) Confusion matrix of MobileNetV2 on test. Takeaway: transfer learning lifts ROC-AUC from 0.68 to 0.97; the remaining errors are mostly false alarms on real videos.

Conclusion. Both MobileNetV2 models meet the recall goal (≥ 0.80) but slightly miss the FPR goal on test (0.217, 0.246; validation 0.17, 0.20). With 138 real test videos the interval includes 0.20; for the final model we will choose the threshold with a margin (validation FPR ≤ 0.15) instead of tuning on test. The GRU is not significantly better, here or on unseen methods (Section 5), so the frame model is our current choice.

Base rate. At a realistic 1% share of fakes, precision would be 0.965·0.01 / (0.965·0.01 + 0.217·0.99) ≈ 4%, about 24 false alarms per real detection: the model is a triage filter for human review, not a blocker. **Cost and latency.** Training takes 0.8 / 1.1 / 7.0 min per epoch on a T4 (B2 / MobileNetV2 / GRU). End to end (notebook 04, 15 FF++ videos of about 13 s, medians), scoring one video takes 2.5 s with the T4 and 2.8 s on the CPU alone. Almost all of it is decoding, face detection and cropping on the CPU (2.5 s); MobileNetV2 needs only 10 ms for its 8 frames on the T4 and 291 ms on the 2-core Colab CPU. Scoring is about 5× faster than real time even without a GPU; face detection, not the model, is the bottleneck.

4. Error Analysis

We inspected the six most confident false alarms and misses of MobileNetV2 on test and of the LOMO run without FaceSwap (Appendix A, Figures 3–4).

- **False alarms are systematic.** The two worst (P(fake) 0.96 and 0.95) are also the two worst in the LOMO-FaceSwap run, a separately trained model. Visible patterns: head covering, glasses, an open mouth with teeth, lowered eyes; all are TV-news footage.
- **Misses depend on the target video, not the method.** All six worst misses are expression reenactments (Face2Face, NeuralTextures), and two target videos are missed by both methods with P(fake) ≤ 0.01. These are calm, frontal speakers with little motion, so the manipulation, and its traces, are small.

Table 3. Error patterns, causes and consequences (MobileNetV2, test).

Example / error type	Likely cause	Security or operational impact	Planned improvement
False positive: real speakers with head covering, glasses, open mouth or lowered eyes (30 of 138 real)	Hypothesis: unusual appearance inside the face crop looks like blending or rendering artifacts; possibly low source quality	Analyst workload (≈ 24 false alarms per true alert at 1% prevalence); a real person can be wrongly discredited; possible unequal treatment of people wearing head coverings or glasses	Slice analysis of FPR by visible attribute (Section 7); threshold with margin; Grad-CAM on these cases
False negative: expression reenactment of calm, frontal talking heads (19 of 544 fakes)	Small manipulation (mouth region only) with few artifacts in a still face	The highest-risk fraud case, a scripted 'CEO' video or video-call clip, is the hardest to detect	Clip-level evidence (GRU, score timeline); mouth-region crops; never treat 'real' as clearance

5. Generalization Testing

Leave-one-method-out (LOMO). To test detection of an unknown attack, we trained MobileNetV2 four times, each time without one manipulation method in training and validation (including the threshold choice), and tested on all methods. Methods seen in training stay at ROC-AUC 0.97–0.99, but the held-out method drops sharply (Figure 2). FaceSwap fakes are even scored as **more real than real videos** (ROC-AUC 0.41), although some show color patches and blending edges a human can see (Appendix A). The model has learned method-specific traces, not general signs of manipulation: knowing FaceSwap lets it catch 90% of Deepfakes, but not the reverse.

Does temporal modeling help with unseen methods? We repeated LOMO with the GRU clip model for the two hardest cases. It does not: for FaceSwap the held-out ROC-AUC falls from 0.41 to 0.34 (paired bootstrap difference −0.075, 95% CI [−0.133, −0.019]); for NeuralTextures it is 0.75 vs. 0.72 (difference +0.025 [−0.024, 0.074], not significant). The GRU's higher NeuralTextures recall at its threshold (0.56 vs. 0.37) comes mostly from a looser operating point (FPR 0.20 vs. 0.06). Like the frame model, the GRU learns the temporal traces of the methods it has seen rather than general flicker.

Table 4. Generalization tests.

Dataset	Differences from training data	Result	Interpretation
Unseen method inside FF++ (LOMO, 4 runs)	One of 4 methods removed from training and validation; 138 real + 136 held-out fakes on test	Held-out recall: Deepfakes 0.90, Face2Face 0.54, NeuralTextures 0.37, FaceSwap 0.13 (ROC-AUC 0.91 / 0.78 / 0.72 / 0.41)	In-distribution scores (≈ 0.97) overstate performance against new tools
Same, GRU clip model (2 runs)	FaceSwap or NeuralTextures held out	Held-out ROC-AUC 0.34 (FaceSwap), 0.75 (NeuralTextures); recall 0.06 / 0.56	Temporal modeling does not close the gap

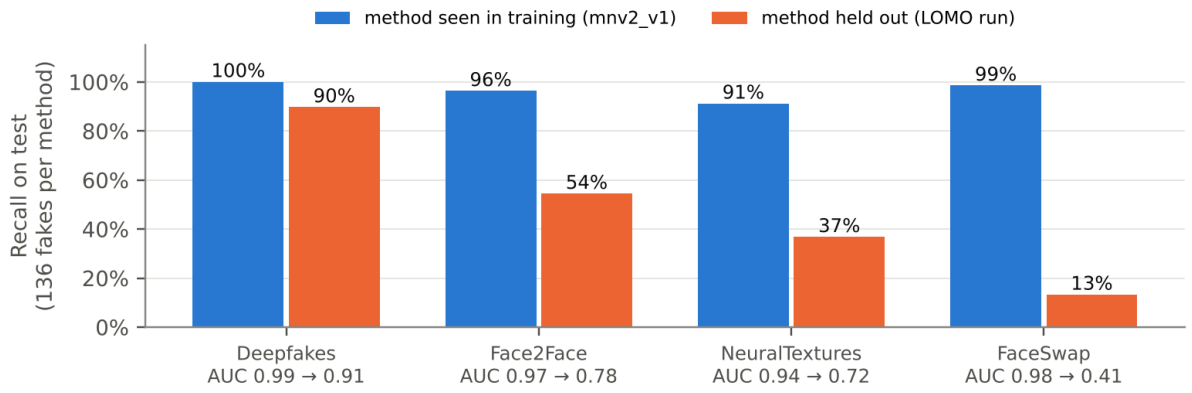


Figure 2. Test recall per method when the method is seen in training (blue) and when it is held out (orange); labels show the ROC-AUC against real videos. Takeaway: detection of unknown manipulation methods is the main weakness.

6. Adversarial Threat Model and Experiment

Scenario. A fraudster sends a manipulated video of an executive or customer (payment request, identity check) and wants the detector to label it real. We assume black-box access: they can try videos and observe the outcome, choose the generation tool, and post-process the file, but cannot see weights or training data.

Preliminary evidence. LOMO already shows the cheapest evasion: switching to a tool the detector has not seen (recall 13–90%). **Planned experiment:** apply realistic post-processing to the test fakes (JPEG re-compression at quality 30–70, downscaling by 0.5, Gaussian blur, Gaussian noise; optionally FGSM as a white-box upper bound) and measure recall at the fixed threshold. **Defense:** fine-tune with these perturbations as augmentation and compare before/after on the same perturbed test set, together with the change in FPR.

Table 5. Threat model.

Asset	Attacker objective	Capability	Attack / test	Defense / mitigation
The detector's decision in a payment or identity-verification workflow (and the analysts' trust in it)	Evasion: a fake classified as real	Black box; unlimited tries; free choice of generator and post-processing; no access to model or data	(1) unseen generator (LOMO, done); (2) compression, resize, blur, noise on test fakes (planned)	Perturbation-augmented fine-tuning; quality gate on input; human review and out-of-band verification for high-value requests

7. Responsible AI, Privacy, and Governance

FF++ has no demographic labels, so we measure unequal performance across manipulation methods (done) and across visible attributes of the real speakers (planned). Privacy rests on the FF++ terms of use: research use only, no videos or frames in the public repository, data kept in a private Google Drive.

Table 6. Responsible-AI checks.

Concern	Test or safeguard	Evidence / current finding	Remaining action
Bias / disparate impact	Recall per manipulation method; FPR on real videos by visible attribute (head covering, glasses, face size, lighting, resolution)	Per method: recall 0.91–1.00 when seen, 0.13–0.90 when unseen. The two worst false alarms involve head covering and glasses (6 cases, not yet evidence)	Annotate the 138 real test videos and report FPR per attribute with intervals by Nov 22
Explainability	Error gallery (done); Grad-CAM on frames and a per-clip score timeline (planned)	Errors are explainable to a reviewer by example; LOMO suggests the model relies on method-specific traces	Grad-CAM for the cases in Figure 3 and the demo
Privacy / governance	FF++ terms of use; only face crops are stored (no full videos); no identities or names kept; output is advisory	Repository holds code and aggregate results only	Model card and data statement in the repository; human review before any action

8. Monitoring, Feedback, and Response Workflow

Response path. Score ≥ threshold → the video goes to an analyst queue with its score timeline and heatmap. For payment or identity decisions, the analyst verifies through a second channel (call-back to a known number); the detector never blocks automatically. Every analyst decision is logged as a label. Because a 'real' verdict is unreliable for new tools (Section 5), it never overrides other fraud controls.

Table 7. Monitoring signals.

Signal / metric	Threshold or trigger	Owner / response	Feedback use
Input drift: face-detection rate, face size, resolution, compression, frame rate	Outside the training range, or weekly share > 2× baseline	ML owner: route out-of-range inputs to manual review	Defines the next data collection
Score drift: share flagged, score distribution (PSI)	PSI > 0.2 or flagged share ± 50% vs. last month	ML owner: check for a new generator or a pipeline change	Triggers a review of the threshold
Alert quality: analyst-confirmed precision; FPR on a fixed set of verified real videos	FPR on the reference set > 0.25	Fraud-analyst lead: raise threshold or pause	Confirmed false alarms become training data
Missed fakes reported after the fact	Any confirmed miss from a new tool	ML owner: add to an evaluation set; retrain with LOMO re-test before redeploying	Quarterly retraining at the latest

9. Final-Phase Plan

Critical path: inference notebook (latency, demo backbone) → robustness experiment and defense → final report. Work is split by the roles from the proposal: Payel on data quality and robustness, Flemming on model evaluation, explainability and the demo.

Table 8. Remaining work.

Remaining task	Owner	Due date	Definition of done
Inference notebook and latency (GPU and CPU)	Flemming Grabowski	Nov 15	ms per video incl. face detection
Robustness experiment + augmentation defense	Payel Sarkar	Nov 22	Recall/FPR before and after per perturbation
Attribute slice analysis of false alarms	Payel Sarkar	Nov 22	FPR per attribute with intervals
Grad-CAM and score timeline	Flemming Grabowski	Nov 22	Figures for report and demo
Demo (upload video → score, timeline, heatmap) + backup recording	Flemming Grabowski	Nov 29	Runs on a laptop; recording saved
Repository: requirements file, tests, model card, README	Payel Sarkar	Nov 29	Fresh Colab run reproduces Table 2
Final report and slides	Both	Dec 4	Submitted Dec 6

References

Cho, K., van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., & Bengio, Y. (2014). Learning phrase representations using RNN encoder–decoder for statistical machine translation. In Proceedings of EMNLP (pp. 1724–1734).

ondyari. (n.d.). FaceForensics++ [GitHub repository], including dataset/splits. Retrieved October 16, 2026, from https://github.com/ondyari/FaceForensics

Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., … Chintala, S. (2019). PyTorch: An imperative style, high-performance deep learning library. In Advances in NeurIPS 32 (pp. 8024–8035).

Rössler, A., Cozzolino, D., Verdoliva, L., Riess, C., Thies, J., & Nießner, M. (2019). FaceForensics++: Learning to detect manipulated facial images. In Proceedings of the IEEE/CVF ICCV (pp. 1–11). https://doi.org/10.1109/ICCV.2019.00009

Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., & Chen, L.-C. (2018). MobileNetV2: Inverted residuals and linear bottlenecks. In Proceedings of the IEEE CVPR (pp. 4510–4520).

Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization. In Proceedings of the IEEE ICCV (pp. 618–626).

Viola, P., & Jones, M. (2001). Rapid object detection using a boosted cascade of simple features. In Proceedings of the IEEE CVPR (pp. I-511–I-518).

Team Contribution Statement

Team member	Responsibilities and completed work	Evidence / artifact
Flemming Grabowski	Data and experiments: downloaded the official FF++ release and ran face extraction (Section 1); ran all training runs in Colab (CNN baseline, MobileNetV2, MobileNetV2 + GRU; 6 LOMO runs) and the test evaluation with bootstrap intervals (Sections 2–5); inspected the error examples (Appendix A); reviewed the full report.	Runs in Drive (runs/*), commit history: https://github.com/payelsarkarr/is665-team6-deepfake-recognition/commits/main
Payel Sarkar	Repository, security and governance: set up and maintained the GitHub repository; wrote the threat model and adversarial-ML plan (Section 6), the responsible-AI, privacy and governance analysis (Section 7) and the monitoring/response workflow and final plan (Sections 8–9); compiled the references; reviewed the full report.	Repository owner (payelsarkarr); Sections 6–9; commit history (link above)

We split the work by area: Flemming owned data preparation, training and evaluation; Payel owned the repository and the security, governance and planning sections. Design decisions (official data only, LOMO, threshold rule) were made together. We coordinated through regular check-ins, this shared document and the shared Drive folder and repository, and each of us reviewed the other's sections before submission. Participation was even, so no adjustment was needed.

AI-Use Disclosure

Required: Disclose any generative-AI tools used for brainstorming, coding, editing, analysis, or slide creation. You remain responsible for the accuracy, originality, security, and reproducibility of all submitted work.

Tool	How it was used	What the team verified or changed

Claude (Anthropic)	Drafted the pipeline code (src/ffpp, notebooks 01–04), proposed the LOMO design, computed bootstrap intervals, and drafted the report.	We ran all experiments ourselves on the official data, checked outputs and figures, and judged the error examples visually. We read, tested and changed the generated code before running it, checked every number in the report against the saved run outputs, and rewrote or shortened drafted text where it overstated results or did not match our decisions.

Submission Checklist

Check	Requirement
✓	All reported results identify the dataset split used.
✓	Baseline and candidate models are compared consistently.
✓	Error analysis includes both causes and security impact.
LOMO only	Generalization testing includes the larger/external dataset.
✓	Threat model identifies attacker capabilities and constraints.
✓	Monitoring and response workflow is actionable.
✓	Contribution statement and AI-use disclosure are complete.

Rubric Self-Check

Criterion	Points	Where the evidence appears
Updated status and project management	1	Snapshot, Section 1
Model development and comparison	2	Section 2, Table 1
Evaluation and error analysis	2	Sections 3–4, Tables 2–3, Figure 1, Appendix A
Generalization testing	2	Section 5, Table 4, Figure 2 (LOMO)
Adversarial ML	1	Section 6, Table 5
Responsible AI, privacy, and governance	1	Section 7, Table 6
Monitoring, response, and final plan	1	Sections 8–9, Tables 7–8

Appendix A. Error Examples

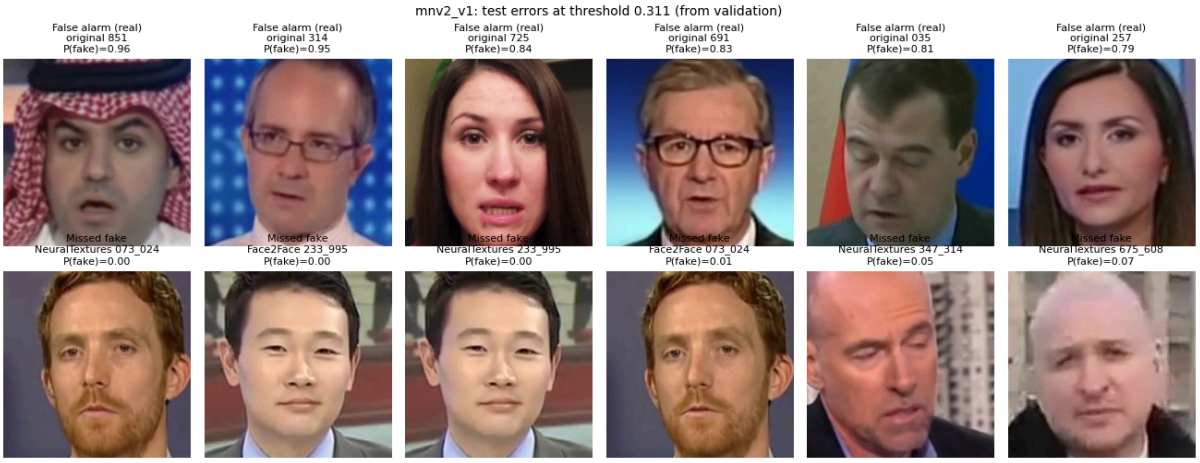


Figure 3. Most confident errors of MobileNetV2 on test (one middle frame per video). Top: real videos flagged as fake. Bottom: fakes passed as real. Takeaway: false alarms share unusual facial appearance; misses are calm, frontal expression edits. Frames from FaceForensics++ (Rössler et al., 2019).

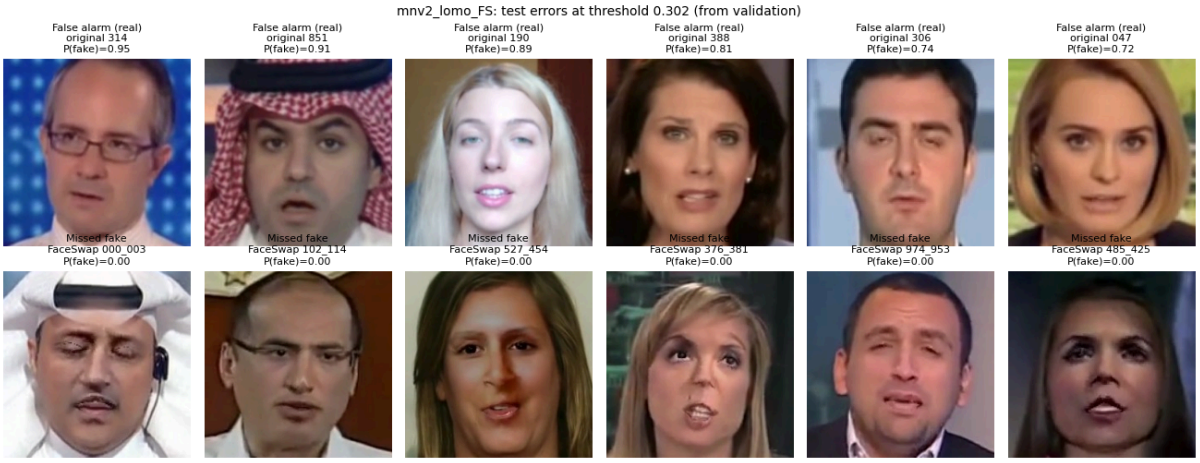


Figure 4. Most confident errors of the MobileNetV2 model trained without FaceSwap (test). Bottom row: FaceSwap fakes scored P(fake) ≈ 0.00 despite visible color patches and blending edges. Top row: the same real videos as in Figure 3 lead the false alarms. Frames from FaceForensics++ (Rössler et al., 2019).